

Dynamic Linear Models and the Kalman Filter

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The Problem

The Model

The Kalman Filter

The Prior

Implementation

Tuning

Summary

Connection to Bayesian inference



Noisy measurements of a moving target

We want to track something that changes over time, but our measurements have error.

Examples:

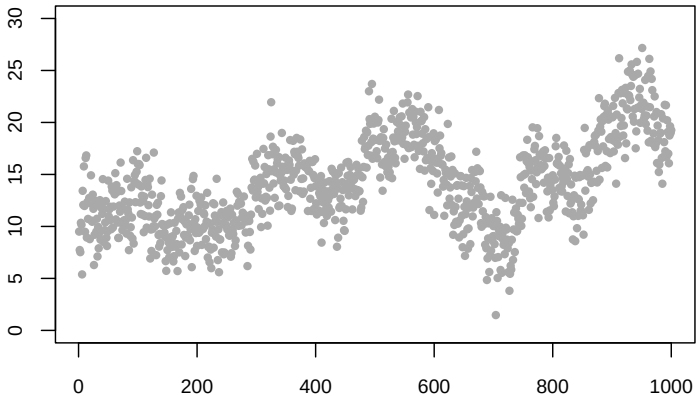
- True unemployment rate (surveys have sampling error)
- Patient's blood pressure (instruments have noise)
- Demand for a product (observed sales are noisy)

Goal: Estimate the true underlying value from noisy data.

Raw data



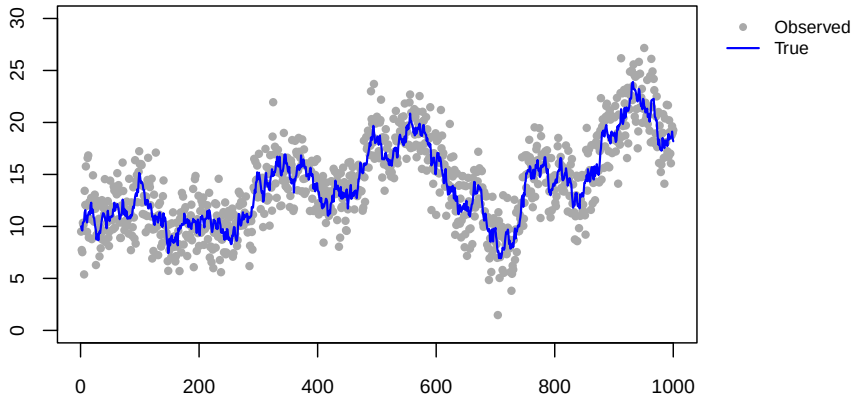
What we observe: noisy measurements over time (gray points).





The true signal

What we *don't* observe: the underlying value θ_t (blue line).



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The model

Let θ_t = true value, Y_t = observation

Observation equation:

$$Y_t = \theta_t + v_t, \quad v_t \sim N(0, V)$$

We see the truth plus measurement noise.

State equation:

$$\theta_t = \theta_{t-1} + w_t, \quad w_t \sim N(0, W)$$

The truth evolves as a random walk.



The two variances

- V = observation noise variance
 - Larger $V \rightarrow$ noisier measurements
- W = state evolution variance
 - Larger $W \rightarrow$ truth can change more

The ratio W/V will determine how the filter behaves.



What we're estimating

The quantity of interest: θ_t = the true underlying value at time t .

- We *observe* Y_t (noisy).
- We *do not observe* θ_t (the signal).
- **Goal:** Estimate θ_t at each time t from the observations $Y_1, \dots, Y_t$.

The Kalman filter gives us an optimal estimate of θ_t (and its uncertainty) at every t .

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The idea

At each time t , combine two pieces of information:

1. **Prediction:** Where we think θ_t is (before seeing Y_t)
2. **Observation:** The new data point Y_t

The filter computes a **weighted average**:

$$\text{Estimate} = (1 - K) \times \text{Prediction} + K \times \text{Observation}$$

where K is the **Kalman gain** (between 0 and 1).



The conditional distributions we care about

At each time t , we care about three distributions:

- **Prior:** $p(\theta_t \mid Y_1, \dots, Y_{t-1})$ (belief about θ_t *before* seeing Y_t)
- **Likelihood:** $p(Y_t \mid \theta_t)$ (how the observation depends on the truth)
- **Target (posterior):** $p(\theta_t \mid Y_1, \dots, Y_t)$ (belief about θ_t *after* seeing Y_t)



How we get the target: Bayes' law

Bayes' law says:

$$p(\theta_t \mid Y_{1:t}) \propto p(\theta_t \mid Y_{1:t-1}) \times p(Y_t \mid \theta_t)$$

$$\text{posterior} \propto \text{prior} \times \text{likelihood}$$

So: combine our prediction (prior) with the new observation (likelihood) to get our updated belief (posterior).



Prior and likelihood from the model

Likelihood (observation equation):

$$Y_t \mid \theta_t \sim N(\theta_t, V)$$

Prior (state equation + last posterior):

$$\theta_t \mid Y_{1:t-1} \sim N(a_t, R_t), \quad a_t = m_{t-1}, \quad R_t = C_{t-1} + W$$

Bayes: posterior $\propto$ prior $\times$ likelihood $\Rightarrow p(\theta_t \mid Y_{1:t})$.



The algorithm

Step 1: PREDICT

$$a_t = m_{t-1} \quad \text{(predicted mean)}$$

$$R_t = C_{t-1} + W \quad \text{(predicted variance)}$$

Step 2: UPDATE

$$K_t = \frac{R_t}{R_t + V} \quad \text{(Kalman gain)}$$

$$m_t = a_t + K_t(Y_t - a_t) \quad \text{(updated mean)}$$

$$C_t = (1 - K_t)R_t \quad \text{(updated variance)}$$



Where do the prediction formulas come from?

Recall the state equation: $\theta_t = \theta_{t-1} + w_t$

Predicted mean:

- Our best guess for θ_{t-1} was m_{t-1}
- The expected change is $E[w_t] = 0$
- So our prediction is: $a_t = m_{t-1} + 0 = m_{t-1}$

Predicted variance:

- We were uncertain about θ_{t-1} by C_{t-1}
- The random walk adds uncertainty W
- Variances add: $R_t = C_{t-1} + W$



Where do the update formulas come from?

We have two pieces of information about θ_t :

- Prediction a_t with variance R_t
- Observation Y_t with variance V

Key idea: When combining estimates, weight by precision (1/variance).

Precision of prediction: $1/R_t$

Precision of observation: $1/V$

Total precision: $1/R_t + 1/V$



Deriving the Kalman gain

Weight on observation = (its precision) / (total precision):

$$K = \frac{1/V}{1/R_t + 1/V}$$

Multiply top and bottom by $R_t \cdot V$:

$$K = \frac{R_t}{R_t + V}$$

$$\text{Weight on prediction} = 1 - K = \frac{V}{R_t + V}$$



Deriving the updated mean and variance

Updated mean (weighted average):

$$m_t = (1 - K) \cdot a_t + K \cdot Y_t = a_t + K(Y_t - a_t)$$

Updated variance (combined precision):

$$\frac{1}{C_t} = \frac{1}{R_t} + \frac{1}{V}$$

Solving for C_t :

$$C_t = \frac{R_t \cdot V}{R_t + V} = R_t \cdot \frac{V}{R_t + V} = R_t(1 - K) = (1 - K)R_t$$



The Kalman gain

$$K_t = \frac{R_t}{R_t + V} = \frac{\text{prediction uncertainty}}{\text{total uncertainty}}$$

- $K \approx 1$: Trust the observation (prediction is uncertain)
- $K \approx 0$: Trust the prediction (observation is noisy)

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Starting the filter

The filter needs a **starting point**:

- m_0 = initial guess for the true value
- C_0 = how uncertain we are about that guess

This is called the **prior**—what we believe before seeing any data.



Choosing the prior

If you have prior knowledge:

- Set m_0 to your best guess
- Set C_0 to reflect your confidence

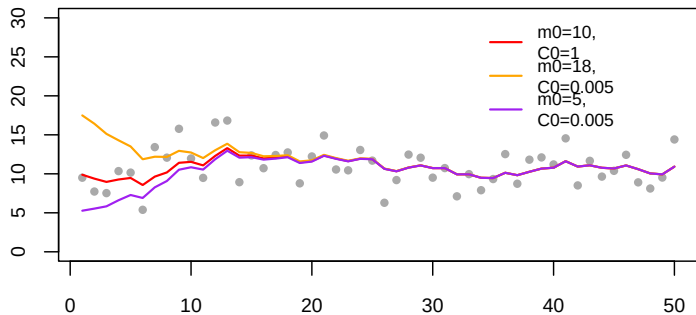
If you don't know:

- Set m_0 to first observation: $m_0 = Y_1$
- Set C_0 large (“diffuse prior”)—says “I’m not sure”

Good news: With enough data, the prior washes out.



Prior doesn't matter much (with enough data)



Different starting points converge to the same estimates (first 50 time steps shown).

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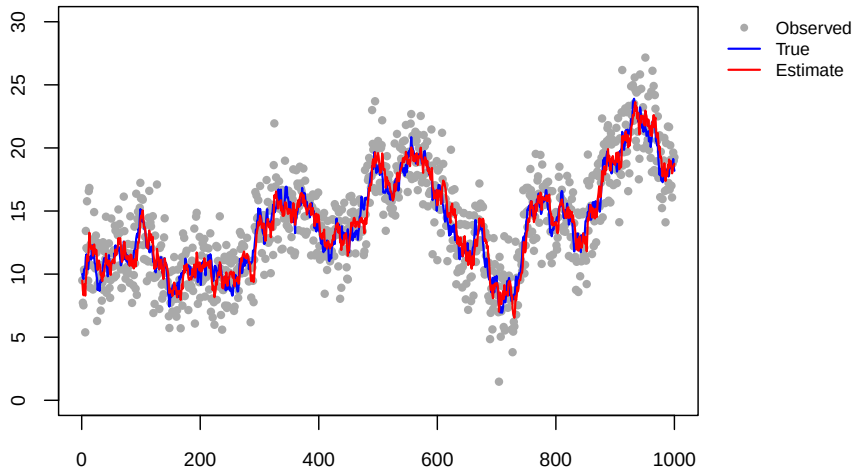
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```
kalman_filter <- function(Y, V, W, m0, C0) {  
  n <- length(Y)  
  m <- C <- numeric(n)  # storage for estimates  
  a <- m0                # initialize  
  
  for (t in 1:n) {  
    # PREDICT:  $R_t = C_{t-1} + W$  (prediction variance)  
    R <- (if (t > 1) C[t-1] else C0) + W  
  
    # UPDATE  
    K <- R / (R + V)  
    m[t] <- a + K * (Y[t] - a)  
    C[t] <- (1 - K) * R  
  
    a <- m[t]  # for next iteration  
  }  
  return(list(m = m, C = C))  
}
```

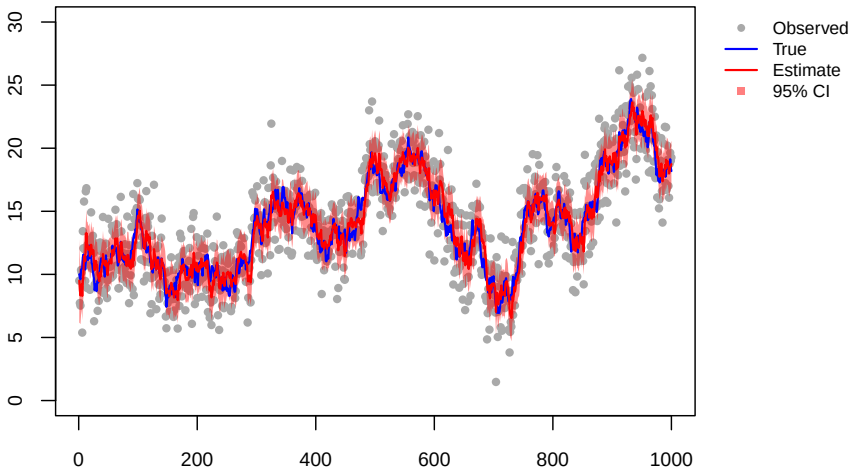
Results



The filter recovers the true signal from noisy observations.



With uncertainty



The filter also tells us how uncertain we are.

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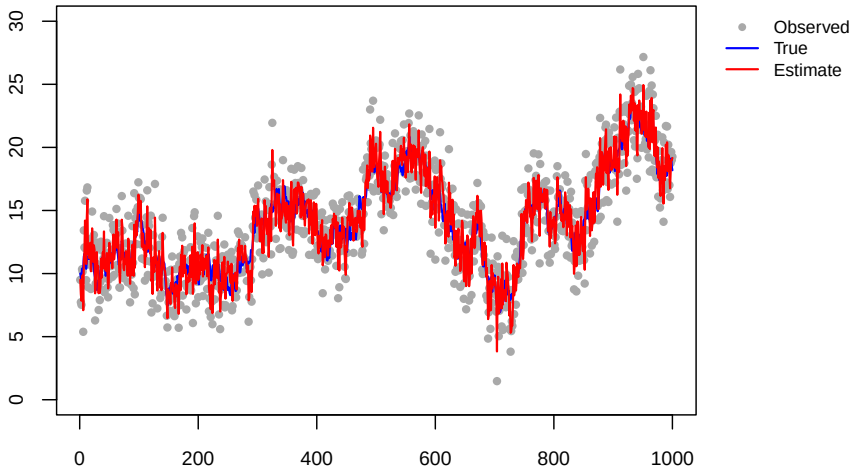
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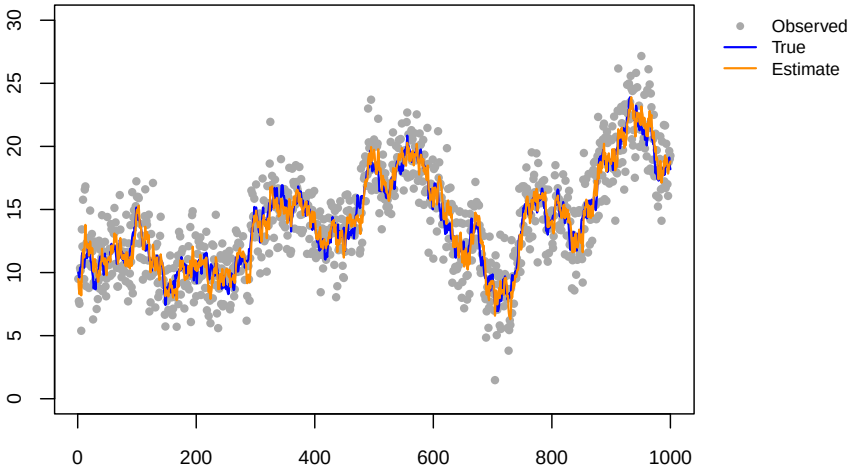
Little smoothing ($W/V = 1$)



Estimate tracks the data closely; little smoothing.



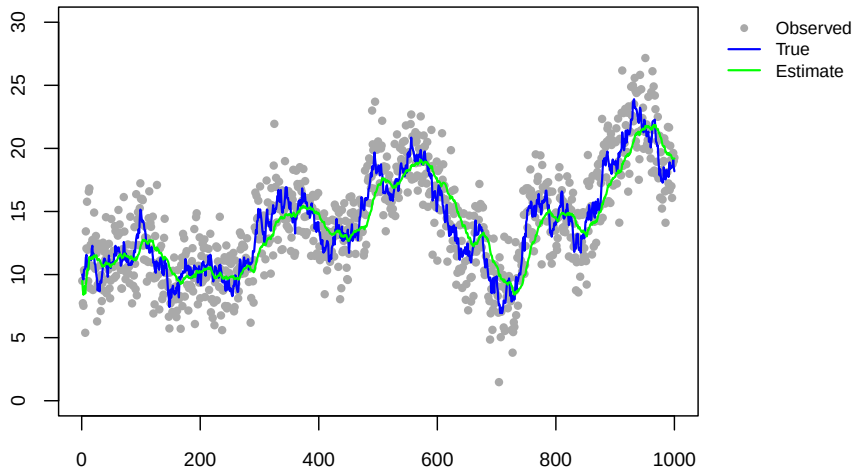
Some smoothing ($W/V = 0.1$)



Moderate smoothing; balance between noise and responsiveness.



Extreme smoothing ($W/V \approx 0.0025$)



Very smooth estimate; slow to react to changes.

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Key points

1. **Model:** Truth is a random walk, observed with noise
2. **Algorithm:** PREDICT $\rightarrow$ UPDATE at each time step
3. **Kalman gain:** Weights prediction vs. observation
4. **Prior:** Starting guess (m_0, C_0) ; washes out with data
5. **Tuning:** W/V ratio controls smoothness



The Kalman filter in one sentence

Optimally combine what you predicted
with what you observed,
weighting by their uncertainties.

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Bayes' law (reminder)

Bayes' law tells us how to update beliefs in light of new data:

$$p(\theta \mid \text{data}) = \frac{p(\theta) p(\text{data} \mid \theta)}{p(\text{data})} \propto p(\theta) \times p(\text{data} \mid \theta)$$

- **Prior** $p(\theta)$: belief about θ *before* seeing the data
- **Likelihood** $p(\text{data} \mid \theta)$: how the data depend on θ
- **Posterior** $p(\theta \mid \text{data})$: belief about θ *after* seeing the data

So: **posterior** $\propto$ **prior** $\times$ **likelihood**.

The **Kalman filter** does exactly this at every time step: the PREDICT step gives the prior for θ_t , the observation Y_t provides the likelihood, and the UPDATE step gives the posterior.



The filter is Bayesian updating

At each time t , the Kalman filter does exactly **Bayesian inference**:

1. **Prior:** Belief about θ_t *before* seeing Y_t
→ from the PREDICT step: $\theta_t \mid Y_{1:t-1} \sim N(a_t, R_t)$
2. **Likelihood:** How the observation Y_t depends on θ_t
→ observation equation: $Y_t \mid \theta_t \sim N(\theta_t, V)$
3. **Posterior:** Belief about θ_t *after* seeing Y_t
→ from the UPDATE step: $\theta_t \mid Y_{1:t} \sim N(m_t, C_t)$

So: **prior** $\times$ **likelihood** $\propto$ **posterior**, at every time step.

Derivation: Prior and likelihood

Prior (from state equation and last posterior):

$$\theta_t \mid Y_{1:t-1} \sim N(a_t, R_t), \quad a_t = m_{t-1}, \quad R_t = C_{t-1} + W$$

So $p(\theta_t \mid Y_{1:t-1}) \propto \exp\left(-\frac{1}{2R_t}(\theta_t - a_t)^2\right)$.

Likelihood (from observation equation):

$$Y_t \mid \theta_t \sim N(\theta_t, V)$$

So $p(Y_t \mid \theta_t) \propto \exp\left(-\frac{1}{2V}(Y_t - \theta_t)^2\right)$.



Derivation: Posterior via Bayes

Fact: Product of two Gaussians (in θ) $\propto$ Gaussian.

Posterior $\propto$ prior $\times$ likelihood gives exponent quadratic in θ_t ;
completing the square yields:

- **Precision:** $\frac{1}{C_t} = \frac{1}{R_t} + \frac{1}{V}$

- **Mean (precision-weighted):**

$$m_t = \frac{(1/R_t) a_t + (1/V) Y_t}{1/R_t + 1/V} = a_t + K_t(Y_t - a_t)$$

Handout: *Posterior via Bayes: Product of Two Gaussians* has the full step-by-step derivation (expand, collect terms, read off C_t and m_t).



Summary: Bayesian view

- The Kalman filter is **recursive Bayesian inference**: at each t , we update from prior to posterior using the new observation.
- Because the prior and likelihood are **Gaussian**, the posterior is Gaussian and the update has a closed form (no MCMC needed).
- The Kalman gain $K_t = R_t / (R_t + V)$ is exactly the weight that Bayes' rule gives to the observation when combining with the prior.

This same idea (prior $\rightarrow$ likelihood $\rightarrow$ posterior) appears throughout Bayesian statistics; the Kalman filter is the dynamic, sequential version for time series.